
EN.625.803 Thesis Progress Report : Continuous-Time Dynamical Systems for Gene Regulatory Networks

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Github Repo: <https://github.com/snjie209/masters>

1 Purpose

This document is an update on the progress made on the first semester course (**EN.625.803**) of the year-long research thesis by Samba Njie for the MS Applied and Computational Mathematics program with the guidance of Professor Thomas Woolf.

2 Context

The circadian molecular clock is a 24-hour process that coordinates the physiology and behavior of living organisms[1]. This internal mechanism is modulated by the gene expression of a diverse set of genes that create a feedback loop, whereby one genes expression may upregulate and downregulate another, in addition to external factors called *zeitgebers* such as light, temperature, etc that also influence the gene expression of these genes[2]. These genes are called core clock genes, and the genes they regulate that directly influence behavior and physiology are clock-controlled genes CCGs[3].

Given the complexity of these interdependent genes, we can treat this as a complex network of interactions and leverage mathematical tools to understand the genetic and external factors that drive the molecular clock over time. In this project, we attempt to use two different methods, Gaussian process regression and neural controlled differential equations[4], to demystify the coupling of these clock-controlled and core clock genes with external factors over time, which may give insight into how they impact mammalian or bacterial circadian physiology.

3 Problem Statement

As such, given a set of genes that regulate the molecular clock and underlying physiology of organisms, we seek to estimate the coupling through probabilistic and signature methods approaches, specifically, via multi-output Gaussian process regression (MOGPR) and neural controlled differential equations (NCDE).

4 Progress Report

4.1 Goals

Below were the planned goals in the first semester:

1. Find a feasible dataset of genes influencing circadian behavior, as well as other extraneous factors that influence the molecular clock, such as light, temperature, etc., that has sufficient number of observations, number of timepoints, and is realistic enough for us to estimate the various mathematical models
2. Find (or write) an implementation for neural controlled differential equations
3. Find (or write) an implementation for multi-output Gaussian process regression

4.2 Progress Summary

Treating the circadian waveforms of gene expression and other external signals as panel data, we have shown in the first semester that both approaches are able to estimate multivariate circadian clock data from realistic single-cell mouse aorta RNA[3] and bulk RNA sequenced gene expression waveforms[5]. The data were derived from Gene Expression Omnibus (GEO)[6], an open source repository that hosts actual sequencing data. Specifically, mouse aorta circadian clock genes were captured using bulk RNA and single-cell RNA sequencing technology to accumulate data sampled over two weeks and 48 hours, respectively, sampling a dozen timepoints [3][5]. Due to the sparsity of data collected, Auerbach et al. (2022)'s hierarchical Bayesian inference algorithm called Tempo was used to resample the data by estimating the amplitude, acrophase, and mesor of these gene expression waveforms[3]. This enabled us to simulate as many realistic waveforms as we wanted to fit to the MOGPR and NCDE models. The result is that we were able to train MOGPRs on a lesser number of waveforms from 24 cycling genes on $T = 100$ time points in a circadian day, and NCDEs estimate the multivariate data on larger number of waveforms ($N = 1000+$).

Below, we delve deeper into each part and our experimental progress:

4.3 Dataset

We have explored various datasets that encapsulate circadian behavior and the genes + zeitgebers associated:

4.3.1 Currently Chosen Dataset

Auerbach et al. (2022) : Mouse bulk RNA + scRNA sequence simulated circadian mouse aorta data

1. This dataset takes in actual scRNA and bulk RNA sequenced data from mouse aorta between scRNA-seq of male mouse aorta collected over 2 weeks and liver hepatocyte data over space and time over thousands of genes. These datasets were used to extract gene expression profiles, which are essentially time series of each gene on sparse intervals of time. A hierarchical Bayesian model was used to estimate the waveform parameters (mesor, amplitude, acrophase, phase).
2. We then took these waveform parameters estimated from real scRNA and bulk RNA sequenced data and fed it into a cosinor function to generate waveforms with larger number of timepoints, enabling us to simulate realistic data to train larger models such as NCDEs on to estimate the coupling between waveforms.

4.3.2 Other Datasets Tried

Hong et al. (2016): Auto-phosphorylation dataset - Cyanobacterial Circadian Clock Metabolite Data

1. Data on metabolites, specifically KaiA, KaiB, and KaiC proteins, which regulate the circadian clock of *Synechococcus elongatus*, as well as other zeitgebers that influence this behavior. There are 20 independent reactions studied observing these metrics, each having 8 observations of the Kai* and zeitgeber concentrations, leading to 169 observations with phosphoforms U, T, D, percent ATP, Kai concentrations, temperature, and time [7]. This was the dataset proposed in the proposal document.
2. The data was not sufficient for training any model, hence we did not proceed with this dataset.

Edwards et al. (2008) CircadiOmics: Arabidopsis circadian clock data

1. The data was found via CircadiOmics, an open-source database of chronobiology gene expression profiles over time for clock-controlled and core clock genes[8]. We explored a plant dataset this time, which involved the *Arabidopsis* plant circadian microarray mRNA data harvested every 4 hours over two days, yielding 13 mRNA samples and 13 chips total.
2. Similar to the above dataset, while this has gene expression data, it does not have enough observations for model training.

Hughes et al. (2008): Mouse liver circadian clock data

1. Another old dataset but with finer granularity as data was collected hourly for 48 hours for 3-5 mice [9].
2. While GPR can train on such data, NCDEs cannot as we need more observations, not only more time points.

4.4 Implementing Signature Method: Neural Controlled Differential Equations

For the signature methods approach by Patrick Kidger (2020) neural controlled differential equations[4], we have tried various packages to implement and tested them on actual data:

Current Methodology: diffrax Package

diffrax is an open-source Python-based library that relies on the Jax framework to solve neural differential equation problems. Jax in itself accelerates numerical computation much like Numpy arrays and enables high-performance numerical computing. With diffrax, we can solve differential equations numerically using well-known solvers, and can even train neural networks with easy-to-use loss functions, gradient generation, and optimization algorithms (in conjunction with optax).

With this framework, we were able to train a neural controlled differential equation with MSE loss, 128-sized hidden layers and 16 neurons for a multi-output regression problem, which trains 24 clock-controlled gene time series, each value being the gene expression for a given timepoint per gene for 100 timepoints and simulated 10K observations (each observation having 24 gene time series) and predicting the next timepoint for each of the 24-genes. We used hyperbolic tangent activation functions in the hidden layers and a linear layer for the output without activation as to not constrain the regressor to a smaller range which does not encapsulate varying gene expression values, and got a small MSE of 0.03 after training for 500 epochs.

We hope to optimize this approach by exploring other neural differential equation implementations or various architectures to ensure we have sufficient accuracy to model the coupling.

4.5 Implementing Gaussian Process regression

Current Methodology: MOGPTK Gaussian Process Regression

Gaussian process regression is a non-parametric, Bayesian inference model that estimates the time series by modeling it as an infinite-dimensional multivariate normal distribution, estimating each infinitesimal time points marginal distribution via a mean function and a clever choice for the covariance function, which smooths the series from discrete to continuous time[10].

Given we have multiple time series due to the various gene expression data over time, we use a specific type of GPR, that is the multi-output Gaussian process regression model, to estimate the structure of multiple outputs (multiple channels) and their dependence amongst each other through the covariance kernel [11]. Instead of simulating $N = 1000+$ waveforms (per gene) as is done with the NCDEs, we sampled only one waveform per gene, and the model then takes each of the 24 genes for 100 timepoints and inverts their covariance matrix to find the covariance across genes and across timepoints and samples the multivariate time series from its posterior distribution. Given only these parameters and an MSE loss, we were able to estimate the time series with less data than the NCDE, albeit with the cost of a much longer computational time.

5 Next Steps

For the second semester, we would still like to expand upon the successes of the first semester as follows:

1. Experiment with different NCDE and MOGPR hyperparameters. For instance, we could replace the loss function of MOGPR and use more efficient solvers for NCDEs.
2. Further explore the coupling between genes using the aforementioned or similar models.
3. Write up the thesis document. We would like to delineate each step of our research process, starting with an introduction of the work, its motivation, the problem statement, context on the biology of circadian clocks, detail each model tried, and the results of our experiments.
4. Present final results in a thesis defense at the end of the quarter.

6 Resources

You can find the latest code in this link (<https://github.com/snjie209/masters>). Please email Samba at snjie1@jhu.edu for any questions or comments.

References

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